

Symbolization Examples and Explanations

This document contains separate sections for Chapters I, II, III, and V, and a special section on problems Symb 3.201-3.225. You can go directly to the material that interests you by using the following hyperlinks.

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Although there is a fairly broad consensus among logic teachers as to what constitutes a correct answer to a symbolization problem, certain fundamental differences between the syntax and vocabulary of natural languages and the syntax and vocabulary of our symbolic language have the effect of making symbolization an inexact science. The program recognizes only one correct answer for most problems. However, it can also determine if you have entered an answer that is tautologically equivalent to a correct answer. Your instructor determines whether to count such answers as correct, “Correct Equivalent”, or incorrect, “Incorrect Equivalent”.

If you believe that you have entered a correct answer that the program does not recognize, please send it us through the Feedback facility.

In order to constrain what is counted as a correct answer, we have adopted (by stipulation) certain conventions, many of which are traditional in logic texts (which is not to say that they are correct by more searching standards). In some, *but not all*, cases a comment will appear along with the problem reminding you of the convention.

Even adopting strict standards of correctness, there may be symbolization problems with acceptable answers that are not recognized by the program. For now, we must depend on your Instructor to warn you of these.

Chapter I Symbolizations – Examples and Explanations

SCHEME OF ABBREVIATION

- P Alfred will pass
S Alfred studies
T Alfred is a student
R Alfred is an administrator

The following are all regarded as stylistic variants of the pedantic form:

It is not the case that Alfred is a student.

$\sim T$

1. Alfred is not a student.
2. Alfred isn't a student.
3. Alfred is other than a student.
4. Alfred fails to be a student. (?)

Although it is questionable whether 4 (and perhaps 3) says nothing more than 1, it is traditional to regard them as stylistic variants of 1, we will make a stipulation for present purposes.

Stipulation 1: For purposes of our class, 1-4 (and analogous lists below) will always be treated as stylistic variants of one another.

The following are all stylistic variants of the pedantic form:

If Alfred is a student, then Alfred studies.

$(T \rightarrow S)$

- | | |
|--|--|
| 5. If Alfred is a student, Alfred studies. | 6. Alfred studies if Alfred is a student. |
| 7. Provided that Alfred is a student, Alfred studies. | 8. Alfred studies provided that Alfred is a student. |
| 9. Given that Alfred is a student, Alfred studies. | 10. Alfred studies given that Alfred is a student. |
| 11. Provided that Alfred is a student, he studies. | 12. Alfred studies provided that he is a student. |
| 13. Given that Alfred is a student, he studies. | 14. Alfred studies given that he is a student. |
| 15. In case Alfred is a student, Alfred studies. | 16. Alfred studies in case Alfred is a student. |
| 17. Assuming that Alfred is a student, Alfred studies. | 18. Alfred studies, assuming that he is a student. |
| 19. On the condition that Alfred is a student, Alfred studies | 20. Alfred studies on the condition that Alfred is a student. |
| 21. Alfred is a student only if Alfred studies. | 22. Only if Alfred studies is Alfred a student. |
| 23. Alfred is a student only on the condition that he studies. | 24. Only on the condition that he studies is Alfred a student. |
| 25. Alfred is a student only given that he studies. | 26. Only given that Alfred studies is he a student. |

Note the following:

The left to right order of the English clauses does not matter. What matters is the clause to which the logical marker – “if”, “only if”, “provided that”, “only on the condition that”, etc. – is attached.

Pronouns, such as “he” may freely replace names, and may even precede the name to which they are linked, as in 24.

The use of “only” has the effect of reversing the role of the logical marker. Whereas “if” normally introduces the logical *antecedent*, “only if” always introduces the logical *consequent*. The same holds the variants of “if”, such as “on the condition that”. The most common error at this stage is to read the “if” as introducing the antecedent while ignoring the “only” as excess verbiage. Think of “only if” as if it were a single hyphenated word, “only-if”.

Thus

$(P \rightarrow Q)$

may appear as

27. If P, Q

28. Q if P

or as

29. P only if Q

30. Only if Q P* [An oddity of this case is that in English, P* must appear in verb-first position, as in 22-26.]

31. P only on the condition Q

32. Only on the condition Q P* [Where P* is the verb-first form of P]]

An understanding of stylistic variants will only help with the structure of simple sentences. The more challenging task is to be able to *parse* a sentence of English so as to reveal **scope**, i.e. which connectives are subordinate to others. For example, does

Alfred will not pass if he is an administrator

have the form

$(R \rightarrow \sim P)$

or the form

$\sim(R \rightarrow P)$

Both seem possible. However,

If Alfred is an administrator, he will not pass.

Can only be symbolized as the first.

When we symbolize, we search for the main connective, and then repeat the process with the components of the main connective. In this way, symbolizing is exactly like parsing a sentence of the symbolic language.

33. Only on the condition that Alfred fails to study will he not pass.

“Only on the condition that” is a variant of “only-if”. So this is a conditional with “Alfred fails to study” as its consequent. (Remember: “only-if” always introduces the consequent.) Thus we have:

33a. (Alfred will not pass $\rightarrow$ Alfred fails to study)

The antecedent is plainly the negation of “Alfred will pass”, and the consequent is plainly the negation of “Alfred studies”. So 33a passes into

33b $(\sim \text{Alfred will pass} \rightarrow \sim \text{Alfred studies})$

“Alfred will pass” and “Alfred studies” are each atomic formulas, that is, they have no logical structure that can be expressed in terms of negation and conditionals. So we look to our *scheme of abbreviation* for their symbolization, and we end up with:

33c $(\sim P \rightarrow \sim S)$

34. Alfred passes if it is not the case that he fails to study, assuming that he isn't an administrator.

In this sentence, both the “if” and the “assuming that” (a variant of “if”) are initial candidates to mark the main break. The two readings might be represented as follows:

Alfred passes if (it is not the case that he fails to study, assuming that he isn't an administrator).
 (Alfred passes if it is not the case that he fails to study), assuming that he isn't an administrator.

Here the presence of the comma resolves the difficulty in favor of the second. Many style manuals have fixed rules for the introduction of commas, but we will, by convention, always regard the presence of a comma as marking a greater break than its absence, and always regard the presence of a semicolon as marking a greater break than a comma.

Stipulation: For purposes of our class, the presence of a comma is always regarded as indicating a greater break than a competing connective without a comma, and a semicolon is always regarded as marking a greater break than a comma.

Thus, since “assuming that”, which is a variant of “if”, introduces the antecedent, we get:

34a $(\text{Alfred isn't an administrator} \rightarrow \text{Alfred passes if it is not the case that he fails to study})$

The consequent is again a conditional with “if” marking the main break, and, as usual, introducing the antecedent. The antecedent of 34a is clearly a negation. So we have:

34b $(\sim \text{Alfred is an administrator} \rightarrow (\text{it is not the case that Alfred fails to study} \rightarrow \text{Alfred passes}))$

Note the replacement of the pronoun by the name. The sentence “it is not the case that Alfred fails to study” is the negation of “Alfred fails to study”, which is itself the negation of “Alfred studies”. So the whole becomes:

34c $(\sim \text{Alfred is an administrator} \rightarrow (\sim \sim \text{Alfred studies} \rightarrow \text{Alfred passes}))$

Here the remaining English components are atomic, that is they are neither negations nor conditionals (atomic is the only other choice in this symbolic language), so we again turn to the scheme of abbreviation:

34c $(\sim R \rightarrow (\sim \sim S \rightarrow P))$

35. Only if if Alfred doesn't study then only given that he isn't an administrator will he pass is he a student.

This string is virtually unreadable, but let's try to show that it is a grammatically correct sentence, or at least that it can be parsed and thus symbolized. Ordinarily we would sprinkle in some commas to help with the reading, but there seems to be only one way to parse it. Let's start by hyphenating the "only if" and its variant so as to avoid getting left with a dangling "only".

35a Only-if if Alfred doesn't study then only-given-that he isn't an administrator will he pass is he a student.

There are four markers of logical structure, "Only-if", "if", "then", and "only-given-that". Although "if" is regularly used without "then", "then" is not normally used (in the sense of a conditional, as opposed to temporal order) without "if". So the "if" and the "then" go together. Since "then" introduces the consequent of a conditional, we must look for a complete sentence following it. Let us place a left parenthesis immediately following "then". Where should the matching right parenthesis go? There is only one possibility since neither:

only-given-that he isn't an administrator

nor

only-given-that he isn't an administrator will he pass is he a student

is a grammatically correct sentence, the sentence introduced by "then" must be

only-given-that he isn't an administrator will he pass

35b Only-if if Alfred doesn't study then (only-given-that he isn't an administrator will he pass) is he a student.

Now that we know what the "then" clause is, we can parenthesize the conditional formed by "if...then..."

35c Only-if (if Alfred doesn't study then (only-given-that he isn't an administrator will he pass)) is he a student.

Because of the difficulty in intuitively grasping the structure of sentence 35, we had to do some work to organize the interior structure. But we could have reached similar results starting on the outside, that is, starting by looking for the main connective and its components. Wherever we start, we must always ascertain that the components of any conditional, or the component of any negation, forms a grammatically correct sentence (possibly in verb-first form for constructions of the form 22, 24, and 26).

The failure of a purported 'antecedent' or 'consequent' to be, by itself, a grammatically correct sentence is a critical sign that the proposed parsing is incorrect.

We can now see that 35c has the overall form of 22. So we have:

35d (Alfred is a student → (if Alfred doesn't study then (only-given-that he isn't an administrator will he pass)))

The consequent of 35d (the material within the blue parentheses) is a pedantic conditional:

35e (Alfred is a student $\rightarrow$ (Alfred doesn't study $\rightarrow$ (only-given-that he isn't an administrator will he pass)))

The consequent of the consequent (the material within the black parentheses) has the form of 26.

Remembering that “only-given-that” introduces the consequent, and replacing some pronouns by names, we obtain:

35f (Alfred is a student $\rightarrow$ (Alfred doesn't study $\rightarrow$ (Alfred will pass $\rightarrow$ Alfred isn't an administrator)))

The antecedent of the consequent is a negation, as is the consequent of the consequent of the consequent, and so, inserting both tildes at once, we have:

35g (Alfred is a student $\rightarrow$ ($\sim$ Alfred does study $\rightarrow$ (Alfred will pass $\rightarrow$ $\sim$ Alfred is an administrator)))

The remaining English components are all atomic. So we use the scheme of abbreviation to obtain:

35h ($T \rightarrow (\sim S \rightarrow (P \rightarrow \sim R))$)

Note that in abbreviating “Alfred does study” by “S”, we implicitly recognize that “Alfred does study” and “Alfred studies” are stylistic variants.

Chapter II Symbolizations – Examples and Explanations

SCHEME OF ABBREVIATION

P	Alfred will pass
S	Alfred studies
T	Alfred is a student
R	Alfred is an administrator
U	Alonzo studies
V	Alonzo is a student
W	Alonzo will pass

The following are all regarded as stylistic variants of the pedantic form:

Alfred is a student, and Alfred studies.

($T \wedge S$)

- | | |
|---|--|
| 1. Alfred is a student, but he studies. | 3. Although he studies, Alfred is a student,. |
| 2. Alfred is a student, although he studies. | 5. Even though Alfred studies, he is a student,. |
| 4. Alfred is a student, even though he studies. | 7. Alfred both is a student and studies. |
| 6. Alfred is both a student and studies. | 9. Alfred is a student that studies. |
| 8. Alfred is a student who studies. | |
| 10. Alfred, who studies, is a student. | |

Here, as in other cases, it is questionable whether 1-5 say no more than 6-10. Although most speakers would not carelessly switch between 1 and 6, it is a matter of genuine controversy in the current literature whether the difference has a bearing on *logical form* and the relation of logical consequence. The preceding list should be understood as reflecting decisions expressed in Stipulation 1 above.

Note that in the pairs 2, 3 and 4, 5 the movement of the dependent clauses, those beginning with “although” and “even though”, to the front does not affect the order of the conjuncts in the symbolized form. Our standard symbolizations follow the practice of symbolizing conjunctions by putting the independent clause on the left and the dependent clause on the right, without reference to which clause appears before the other in the English.

The following are all regarded as stylistic variants of the pedantic form:

Alfred is a student or Alfred is an administrator.

($T \vee R$)

- | | |
|--|--|
| 11. Alfred is a student or Alfred is an administrator. | 12. Either Alfred is a student or Alfred is an administrator. |
| 13. Alfred is a student unless Alfred is an administrator. | 14. Unless Alfred is an administrator, he is a student. |
| 15. Alfred is a student or an administrator. | 16. Alfred is either a student or an administrator. |
| 17. Alfred either is a student or is an administrator. | 17a. Alfred is a student except in the case that he is an administrator. |

In English “or” is said to have two senses, an “inclusive” sense, in which it means *either or both* sometimes written as “and/or”, and an exclusive sense in which it means *one or the other but not both*. Many speakers hear “unless” as exclusive “or” as in “He will pass unless he doesn’t study (in which case he won’t)” with the parenthetical being understood. The same holds for exception-clauses in general, such as “P except in the case that Q”. Both “unless” and “except in the case that” may be used in the sense of *exclusive* “or”, but our convention is to always use *inclusive* “or”. In order to resolve these difficulties for present purposes, we will make a stipulation.

Stipulation 2: For purposes of our class, “or”, “unless”, and exception clauses will always be treated as occurring in the inclusive sense, and thus symbolized using “ $\vee$ ”.

Note once again that in the pair 13, 14 the movement of the dependent clause beginning with “unless” to the front does not affect the order of the disjuncts in the symbolized form. Our standard symbolizations continue to follow the practice of putting the independent clause on the left and the dependent clause on the right, without reference to which clause appears before the other in the English.

The following are all regarded as stylistic variants of the pedantic form:

Alfred is a student if and only if Alfred studies.

($T \leftrightarrow S$)

18. Alfred is a student exactly on the condition that he studies.
 19. Alfred is a student just in case he studies. 20. Just in case Alfred studies is he a student.

One other idiom, “Neither/nor”, requires special attention. Logicians commonly claim that either of two symbolizations is adequate for,

It is neither the case that P nor that Q

The two symbolizations are:

$\sim(P \vee Q)$

$(\sim P \wedge \sim Q)$

Since “neither/nor” seems so plainly to have resulted from joining a negative particle with “either/or” we favor the first. In view of the fact that the two are tautologically equivalent, the second will be regarded as a Correct Equivalent, but the standard symbolization answers will favor the first.

A primary task of symbolization is to unwind complex subjects and predicates into complex combinations of sentences. Almost unlimited combinations of “and”, “or”, and “not” (and their variants) are possible in complex subjects and predicates, for example,

21. Either Alfred but not Alonzo or Alonzo but not Alfred attended.

which unwinds to

22. Either Alfred attended and Alfred did not attend or Alonzo attended and Alfred did not attend.

Since both subject and predicate may be complex, we are faced with sentences like,

23. Alfred and Alonzo either jumped or were pushed.

Here the issue arises whether the conjunction implicit in the complex subject or the disjunction implicit in the complex predicate has primary scope. We will resolve this issue with a stipulation (which is thought to be correct, not merely traditional).

Stipulation 3: In cases in which both the subject and predicate are complex, we will regard the subject as taking scope over the predicate.

According to the foregoing stipulation, 23 will be treated as equivalent to

23a. Alfred either jumped or was pushed, and Alonzo either jumped or was pushed.
(note that the comma makes the “and” the main connective)

and this in turn is equivalent to

23b. Either Alfred jumped or Alfred was pushed, and either Alonzo jumped or Alonzo was pushed.

Ambiguities can arise from the mixing of “and” and “or”.

In the absence of punctuation, a mix of “and”s and “or”s can be ambiguous as to which has greater scope. Consider

24. Alfred will study and pass or be frustrated.

This is ambiguous between

25. Alfred will (study and pass) or be frustrated.

26. Alfred will study and (pass or be frustrated).

which are *not* equivalent to one another. In such cases, the insertion of “either” or “both” can often be used to disambiguate, since the material between “either” and its matching “or” must form the first disjunct of the disjunction, and the material between “both” and its matching “and” must form the first conjunct of the conjunction. 25 and 26 can be rendered unambiguously (and without parentheses) as,

25a. Alfred will either study and pass or be frustrated.

26a. Alfred will study and either pass or be frustrated.

In determining what to count as a correct symbolization, all explicit indications of grouping, including commas and semicolons, and the placement of “either” and “both” are followed scrupulously.

For example, “P and R or S” would be treated as ambiguous. But “P and R, or S”, and “P and either R or S” would not be.

There is no uniquely “correct” symbolization of conjunctions and disjunctions with more than two components, but we will follow a convention.

In our symbolism, “ $\wedge$ ” and “ $\vee$ ” are two-place connectives, but of course English allows us to make conjunctions and disjunctions as long as we wish, as in “Rudolf, Alonzo, Kurt, and Van will either phone, fax,

or email their current addresses.” This difference between English syntax and the syntax of our symbolic language requires us to make an arbitrary decision as to how to symbolize a series of conjuncts or disjuncts. We will treat them as if associated to the left. So a series such as “P and Q and R and S” will be symbolized, in official notation, as “ $((P \wedge Q) \wedge R) \wedge S$ ”. Note that it is only when associated in this way that parentheses can be dropped to form “ $(P \wedge Q \wedge R \wedge S)$ ”. As with the two symbolizations of “neither/nor”, other groupings of a series of conjuncts or disjuncts will be tautologically equivalent to our standard treatment, and thus should be marked as Correct Equivalents. But the standard symbolization answers to such series will be the one that causes the tree to branch to the left.

Chapter III Symbolizations – Examples and Explanations

In each of the following groups, all of the English sentences are regarded as stylistic variants of one another. Their symbolization is found at the bottom of each group. Most of the basic English forms that you are likely to encounter in a symbolization problem should be found here.

The following section contains a detailed analysis of the complex forms found in the symbolization problems that begin with 3.201.

F{1}: {1} is a student
G{1}: {1} studies
H{1}: {1} is an administrator
I{1}: {1} is a teacher
a: Alonzo

Please note that we use “{1}” as a placeholder, where our Textbook uses the variable “x”.

5. Everything is a student.
6. Each thing is a student.
7. All things are students.

$\forall xFx$

8. Something is a student.
9. At least one thing is a student.
10. There is a student.

$\exists xFx$

11. Some student studies.
12. Some students study.
13. At least one student studies.
14. There is a student who studies.
15. A (certain) student studies.

$\exists x(Fx \wedge Gx)$

16. Some student doesn't study.
17. Some students don't study.

$\exists x(Fx \wedge \sim Gx)$

18. All students study.
19. Each student studies.
20. Every student studies.
21. Everything that is a student studies.
22. Anything that is a student studies.
23. A (generic) student studies.
24. Any student studies.
25. If anything is a student, it studies.

$\forall x(Fx \rightarrow Gx)$

26. Only students study.
 27. None but students study.
 Nothing other than students studies
 $\forall x(Gx \rightarrow Fx)$
 $\sim \exists x(\sim Fx \wedge Gx)$
28. All and only students study.
 $\forall x(Gx \leftrightarrow Fx)$
 $\forall x(Fx \rightarrow Gx) \wedge \forall x(Gx \rightarrow Fx)$
29. No students study.
 $\sim \exists x(Fx \wedge Gx)$
30. Students don't study.
 $\forall x(Fx \rightarrow \sim Gx)$
31. All students and teachers study.
 $\forall x(Fx \rightarrow Gx) \wedge \forall x(Ix \rightarrow Gx)$
32. All teachers are students and study.
 $\forall x(Ix \rightarrow Fx \wedge Gx)$
33. No teacher is a student unless he or she studies.
 $\forall x(Ix \rightarrow (\sim Fx \vee Gx))$
 $\sim \exists x(Ix \wedge Fx \wedge \sim Gx)$
34. No teacher is a student unless Alonzo studies.
 $\forall x(Ix \rightarrow (\sim Fx \vee Ga))$
 $\forall x(Ix \rightarrow \sim Fx) \vee Ga$
 $\sim \exists x(Ix \wedge Fx \wedge \sim Ga)$
 $\sim \exists x(Ix \wedge Fx) \vee Ga$
35. Among administrators only students and teachers study.
 $\forall x(Hx \rightarrow (Gx \rightarrow Fx \vee Ix))$
36. Teachers who study are students.
 $\forall x(Ix \wedge Gx \rightarrow Fx)$
37. Teachers, who study, are students.
 $\forall x(Ix \rightarrow Fx) \wedge \forall x(Ix \rightarrow Gx)$
38. If every teacher studies, any student studies.
 $\forall x(Ix \rightarrow Gx) \rightarrow \forall x(Fx \rightarrow Gx)$
39. If any student studies, every teacher studies.
 $\exists x(Fx \wedge Gx) \rightarrow \forall x(Ix \rightarrow Gx)$

40. If Alonzo studies, any student studies.

$$Ga \rightarrow \forall x(Fx \rightarrow Gx)$$

41. If any student studies, Alonzo studies.

$$\exists x(Fx \wedge Gx) \rightarrow Ga$$

42. If any student studies, he or she is an administrator.

$$\forall x(Fx \wedge Gx \rightarrow Hx)$$

Persons, personalities, and personal pronouns.

Words like “who” and “whom” are *personal* pronouns. As used in “someone” and “everyone”, the pronoun “one” is also a personal pronoun. Personal pronouns carry with them the claim (or presupposition) that they refer to persons in just the way that “he” and “she” carry claims (or presuppositions) about gender. Jacob Mojarro notes that we sometimes use personal pronouns with animals other than persons, indeed with anything that has *personality*. “I have a dog who loves to swim.” So perhaps they should be called *personality pronouns*. In any case, we will adopt a stipulation that personal pronouns carry a claim of personhood.

Hence,

There is a student that studies.

$$\exists x(x \text{ is a student} \wedge x \text{ studies})$$

There is a student *who* studies.

$$\exists x(x \text{ is a student} \wedge x \text{ is a person} \wedge x \text{ studies})$$

Everything that studies passes.

$$\forall x(x \text{ studies} \rightarrow x \text{ passes})$$

Everyone that studies passes.

$$\forall x(x \text{ studies} \wedge x \text{ is a person} \rightarrow x \text{ passes})$$

Stipulation 4: The use of a personal pronoun will be taken to imply a restriction to persons.

“Any” and “a” as determiners.

As noted above (and in 40-41, and in 15 and 23), “any” and “a” can stand for either universal or existential quantifiers, depending on context. These determiners are especially sensitive to negation. Note that “John can pass any course” is synonymous with “John can pass every course”; whereas “John cannot pass any course” is synonymous with “John cannot pass even one [at least one] course”. For this reason, it is useful when parsing to try and transform “any” and “a” into either “every” or “some” at an early opportunity. Otherwise, there is a risk of misreading which quantifier the “any” or the “a” stands for. For example, if in parsing “John cannot pass any course”, one first determined it to be a negation, and wrote the negated formula as “John can pass any course” (rather than “John can pass some course”), one would likely end up with the incorrect:

$$\sim \forall x(x \text{ is a course} \rightarrow \text{John can pass } x)$$

rather than the correct:

$$\sim \exists x(x \text{ is a course} \wedge \text{John can pass } x)$$

Discussion of Symbolization Problems

3.201 – 3.225

Symbolize the following English sentences using the given scheme of abbreviation.

F{1}: {1} jumps	I{1}: {1} is a dog	L{1}: {1} is yellow
G{1}: {1} sings	J{1}: {1} is a canary	M{1}: {1} is long-haired
H{1}: {1} is petted	K{1}: {1} is a dolphin	N{1}: {1} is a pet
O{1}: {1} can fly		
a: Spot	b: Kiwi	c: Feathers

3.201 Dogs and dolphins jump if petted.

This sentence, whose subject contains an English conjunction is treated as a sentential conjunction. Note that the subject is not “Dogs who are also dolphins” or “Dolphin-dogs”, which are not conjunctions in English but rather relative clause or adjectival constructions. (1) is analogous to, “Spot and Kiwi jump if petted.”

The conjunction is,

“dogs jump if petted and dolphins jump if petted”

Consider the first conjunct,

“dogs jump if petted”

This is a (plural) subject/predicate sentence with a complex predicate. It is parsed as,

“[dogs] [jump if petted]”

(Note that the parsing cannot be “[dogs jump] if [petted]” with “if” as the main connective, because the predicate “petted” does not form a sentence by itself. Such a parsing leaves the predicate stranded, without a subject.)

A subject/predicate sentence in which the subject is a ‘bare’ plural (i.e. a plural, like “dogs” or “small dogs”, but lacking a determiner, such as “all” or “some”) is traditionally symbolized with a ‘restricted’ universal quantifier. (Note, however, that there are cases in which the bare plural is existential, e.g. “UCLA students painted the statue of Tommy Trojan blue and gold.”) A ‘restricted’ universal quantifier is a universal quantifier over a conditional, whose antecedent restricts the values of the variable to the subject, and whose consequent holds the predicate:

$\forall x(x \text{ is a dog} \rightarrow x \text{ jumps if petted})$

and ultimately

$\forall x(Ix \rightarrow (Hx \rightarrow Fx))$

The whole conjunction then becomes,

$\forall x(Ix \rightarrow (Hx \rightarrow Fx)) \wedge \forall x(Kx \rightarrow (Hx \rightarrow Fx))$

3.202 Only dogs and dolphins jump if petted.

“Only” is somewhat deviant from other expressions used as determiners. It is often useful to paraphrase “only” as “nothing but”, “nothing other than” or “nothing distinct from”. This paraphrase, along with an adjustment of “jump” to make it singular, yields,

“nothing distinct from (both) dogs and dolphins jumps if petted”.

Since “no F” is traditionally read as “not some F”, this is easily read as,

“~something distinct from (both) dogs and dolphins jumps if petted”.

The negated English sentence has the determiner phrase “something distinct from (both) dogs and dolphins” as subject, and “jumps if petted” as predicate. Such sentences are traditionally symbolized as restricted existential generalizations (see 5 below) yielding,

$\sim \exists x(x \text{ is distinct from both dogs and dolphins} \wedge x \text{ jumps if petted})$.

From here, there are no special problems. The left conjunct is a stylistic variant of “x is not a dog and not a dolphin”, and the right conjunct a stylistic variant of “if x is petted, it jumps”, so the whole is symbolized as,

$$\sim \exists x((\sim Ix \wedge \sim Kx) \wedge (Hx \rightarrow Fx))$$

This suggestion differs from the text, which treat “only F” like “all F” except that the conditional is reversed. But there doesn't seem a way to apply that algorithm here. “All dogs and dolphins are G” is a conjunction: “All dogs are G and all dolphins are G”, so is “Only dogs and dolphins are G” supposed to be “Only dogs are G and only dolphins are G”? This is manifestly wrong. As is “All Gs are dogs and dolphins”. So it is hard to see a way to apply the “reverse the conditional” idea to this case. On the other hand, a few simple operations (including QN) quickly give us “Everything that jumps if petted is either a dog or a dolphin”, and all these QN-variants probably should be treated as acceptable, if not as literal as might be.

3.203 Dogs and dolphins jump only if petted.

The analysis is exactly as for 1, except that the predicate “jumps only if petted” is symbolized,

“(x jumps $\rightarrow$ x is petted)”

So the whole conjunction is symbolized,

$$\forall x(Ix \rightarrow (Fx \rightarrow Hx)) \wedge \forall x(Kx \rightarrow (Fx \rightarrow Hx))$$

3.204 Dogs and dolphins jump only if Spot and Kiwi are petted.

Like 1, this sentence is a conjunction. Consider the first conjunct,

“dogs jump only if Spot and Kiwi are petted”

I think that the most natural reading gives it the overall form of a plural subject/predicate sentence, like “dogs jump only if petted”. This means parsing it as follows,

“[Dogs] [jump only if Spot and Kiwi are petted]”

Subject/predicate sentences in which the subject is a ‘bare’ plural are discussed in 1.

The traditional ‘restricted’ universal quantifier symbolization would be,

$\forall x(x \text{ is a dog} \rightarrow x \text{ jumps only if Spot and Kiwi are petted})$

and ultimately

$\forall x(Ix \rightarrow (Fx \rightarrow Ha \wedge Hb))$

This is logically equivalent (by Importation and Confinement) to

$\exists x(Ix \wedge Fx) \rightarrow Ha \wedge Hb$

“If some dog jumps, then Spot and Kiwi are petted.”

The whole conjunction then becomes,

$\forall x(Ix \rightarrow (Fx \rightarrow Ha \wedge Hb)) \wedge \forall x(Kx \rightarrow (Fx \rightarrow Ha \wedge Hb))$

This is logically equivalent to,

$(\exists x(Ix \wedge Fx) \vee \exists x(Kx \wedge Fx)) \rightarrow Ha \wedge Hb$

An alternative, non-equivalent, reading of

“Dogs jump only if Spot and Kiwi are petted”

is to regard the “only if” as marking the major break, parsing as follows,

“[Dogs jump] only if [Spot and Kiwi are petted]”,

and symbolizing as

$(\text{Dogs jump} \rightarrow \text{Spot and Kiwi are petted})$.

If the antecedent, whose subject is a bare plural, is regarded as a restricted universal, it becomes

$\forall x(Ix \rightarrow Fx) \rightarrow Ha \wedge Hb$

“If every dog jumps, then Spot and Kiwi are petted.”

The whole conjunction then becomes,

$(\forall x(Ix \rightarrow Fx) \rightarrow Ha \wedge Hb) \wedge (\forall x(Kx \rightarrow Fx) \rightarrow Ha \wedge Hb)$

This is logically equivalent to,

$(\forall x(Ix \rightarrow Fx) \vee \forall x(Kx \rightarrow Fx)) \rightarrow Ha \wedge Hb$

A third reading, which I think implausible, makes “only if” the main connective of the original sentence. It gives us,

$(\forall x(Ix \rightarrow Fx) \wedge \forall x(Kx \rightarrow Fx)) \rightarrow Ha \wedge Hb$

Which reading do you think is correct?

3.205 No pet neither sings nor jumps.

Here the subject is a familiar determiner phrase (determiner “no” plus noun “pet”), and the predicate is “neither sings nor jumps”. Sentences of the form “no F is G” are traditionally regarded as stylistic variants of negations of restricted existential generalizations, i.e. as having the form,

“it is not the case that some F is G”,

in which,

“some F is G”

is the ‘restricted’ existential generalization. A ‘restricted’ existential quantifier is a existential quantifier over a conjunction, whose left conjunct restricts the values of the variable to the subject, and whose right conjunct holds the predicate:

$$\exists x(Fx \wedge Gx)$$

In the present case the restriction is to pets, and the predicate is a complex neither-nor construction,

$$\exists x(x \text{ is a pet} \wedge x \text{ neither sings nor jumps})$$

So the symbolization of the whole sentence becomes,

$$\sim \exists x(Nx \wedge \sim(Gx \vee Fx))$$

The text and others suggest alternative symbolizations of “neither P nor Q”, for example as “not P and not Q”, but we have agreed on the more literal reading “not (P or Q)”.

3.206 Some dogs both sing and jump, but only long-haired dogs are petted.

I think that the most natural reading of the second conjunct

“Only long-haired dogs are petted”

is as a stylistic variant of

“Among dogs, only the long-haired are petted.”,

which is symbolized

$$\forall x(Ix \rightarrow (Hx \rightarrow Mx)).$$

Thus the symbolization of the whole original sentence becomes,

$$\exists x(Ix \wedge (Gx \wedge Fx)) \wedge \forall x(Ix \rightarrow (Hx \rightarrow Mx))$$

An alternative, non-equivalent, reading of the second conjunct is as a stylistic variant of

“nothing that is not a long-haired dog is petted”

or perhaps the equivalent

“whatever is petted is a long-haired dog”,

which is symbolized

$$\forall x(Hx \rightarrow (Mx \wedge Ix)).$$

On this alternative reading, the symbolization of the whole original sentence becomes,

$$\exists x(Ix \wedge (Gx \wedge Fx)) \wedge \forall x(Hx \rightarrow (Mx \wedge Ix))$$

Which reading do you think is correct?

3.207 If a pet is yellow, then it sings but doesn't jump.

Because the sentence begins with "if", some may think its overall form to be a conditional,

"If [a pet is yellow], then [it sings but doesn't jump]".

However, this would leave the pronoun "it" stranded in the consequent, outside the scope of the determiner phrase "a pet" to which it is bound.

(Note that a pronoun can be a mere shorthand for its antecedent rather than representing a variable 'bound' to its antecedent. The "he" in "if Spot sings, he jumps" can be viewed as a mere shorthand. The result of replacing "he" by "Spot" is equivalent to the original. But the "it" in "if a pet is yellow, it sings" functions differently. The result of replacing "it" by "a pet" says something much weaker. It is not equivalent to the original.)

Had sentence 7 been,

"If [any pet is yellow], then [Feathers sings but doesn't jump]",

the conditional analysis would have been appropriate because there would have been no problem of stranded pronouns. So 7 should be treated as a stylistic variant of,

"a pet is yellow only if it sings but doesn't jump".

We encountered this form in 3 and within the conjuncts of 4, but with bare plurals rather than with a determiner phrase like "a pet". It should be parsed as having subject/predicate form,

"[a pet] [is yellow only if it sings but doesn't jump]".

The determiner phrases "a pet" and "any pet" can appear as a universal ("a [any] teacher must prepare") or as an existential ("a teacher broke the copy machine"). But in this case, the form

"if an F is G, then it is H",

requires the universal interpretation. It is thus symbolized as a restricted universal generalization,

$\forall x(x \text{ is a pet} \rightarrow x \text{ is yellow only if it sings but doesn't jump})$

The consequent raises an interesting question of scope. Is it ambiguous as between whether "only if" or "but" is the main connective? No! The word "but" cannot be the main connective, for then the predicate "doesn't jump" would be stranded, without a subject. So "only if" must be the main connective, and the symbolization of the whole sentence becomes,

$\forall x(Nx \rightarrow (Lx \rightarrow Gx \wedge \sim Fx))$

3.208 If Feathers sings and all who sing jump, then Feathers will not fail to jump.

Here the "If... then ..." structure is not misleading. There is no pronoun in the "then" clause that represents a variable 'bound' to a determiner phrase in the "if" clause. So the main structure is,

(Feathers sings and all who sing jump $\rightarrow$ Feathers will not fail to jump),

and the symbolization of the whole sentence becomes,

$(Gc \wedge \forall x(Gx \rightarrow Fx) \rightarrow \sim \sim Fc)$

3.209 Given that all pets jump or sing and no canary jumps, no canary is a pet unless it sings.

The overall structure is sentential,

(all pets jump or sing $\wedge$ no canary jumps $\rightarrow$ no canary is a pet unless it sings)

We are already familiar with the “all F are G” form (restricted universal generalization) and the “no F is G” form (negation of a restricted existential generalization) so they should present no difficulty,

$[\forall x(x \text{ is a pet} \rightarrow x \text{ jumps or sings}) \wedge \sim \exists x(x \text{ is a canary} \wedge x \text{ jumps}) \rightarrow$

no canary is a pet unless it sings)]

The consequent is of the puzzling “no/unless” form: “no F is G unless it is H”. One wants to get both negation and disjunction into such a sentence. But if we were to treat it as a variety of “no F is G”, namely as having the form “ $\sim \exists x(x \text{ is a canary} \wedge x \text{ is a pet unless it sings})$ ” --which is already a doubtful paraphrase -- the “unless” would have to be treated as synonymous with “and not” rather than “or”,

$\sim \exists x(Jx \wedge (Nx \wedge \sim Gx))$

(Note that treating “unless” as “or” would yield “ $\sim \exists x(Jx \wedge (Nx \vee Gx))$ ”, which is logically equivalent to “ $\sim \exists x(Jx \wedge Nx) \wedge \sim \exists x(Jx \wedge Gx)$ ”. Can you derive each from the other? The latter says that no canary is a pet and no canary sings, which is certainly not what 9 says.)

So if we wish to symbolize “unless” as a disjunction, we must give up the usual symbolization of “no F”. The best I can do is,

$\forall x(Jx \rightarrow (\sim Nx \vee Gx))$.

Either symbolization of “no canary is a pet unless it sings” is acceptable. Can you derive each from the other? I usually use the first because the “no F” determiner phrase seems more stable than the “unless”. So I would symbolize the whole sentence as,

$[\forall x(Nx \rightarrow Fx \vee Gx) \wedge \sim \exists x(Jx \wedge Fx) \rightarrow \sim \exists x(Jx \wedge (Nx \wedge \sim Gx))]$

but

$[\forall x(Nx \rightarrow Fx \vee Gx) \wedge \sim \exists x(Jx \wedge Fx) \rightarrow \forall x(Jx \rightarrow (\sim Nx \vee Gx))]$

is OK too.

By the way, 209 sounds like a logical truth. Can you prove it?

3.210 No dolphin sings unless it jumps.

“No/unless” sentences of this form are discussed under 209 above. Either of two symbolizations is acceptable,

$\sim \exists x(Kx \wedge (Gx \wedge \sim Fx))$

or

$\forall x(Kx \rightarrow (\sim Gx \vee Fx))$

3.211 No dolphin sings unless Kiwi jumps.

Because there is no pronoun in the “unless” clause that is bound to the determiner phrase “no dolphin”, this is a different sort of “no/unless” sentence, one that is amenable to regarding “unless” as the main connective, and then treating “no dolphin sings” in the traditional way described in 5,

$\sim \exists x(Kx \wedge Gx) \vee Fb$

Question: If the “Fx” in either symbolization of 10 were replaced by “FB”, would the result be logically equivalent to the above?

3.212 If a dolphin is petted, it will jump.

This is a simpler variant of 7. The use of the future tense in the consequent may be thought to cast doubt on the symbolization, but such sentences are traditionally symbolized with an eye to what is being said rather than to the exact tense structure of the sentence,

$$\forall x(Kx \rightarrow (Hx \rightarrow Fx))$$

A less literal interpretation of 12 might regard it as a stylistic of,
 “a dolphin that is petted will sing”.

The latter is also a restricted universal generalization, but here the restrictor includes the relative clause,

$$\forall x(Kx \wedge Hx \rightarrow Fx)$$

These two are logically equivalent by the law of exportation/importation.

3.213 If all dolphins are petted, they will all jump.

The point of this problem is to illustrate a case in which a plural pronoun is a mere shorthand for its antecedent rather than representing a variable that is bound to an antecedent determiner phrase (see the parenthetical note in 7). The plural pronoun “they” is merely a shorthand. Thus, 13 is a stylistic variant of,

“If all dolphins are petted, all dolphins will jump”,
 a conditional with restricted universal generalizations in both its antecedent and consequent.

$$\forall x(Kx \rightarrow Hx) \rightarrow \forall x(Kx \rightarrow Fx)$$

3.214 A dolphin jumps. [ambiguous?]

As noted in 7, when used with the present and future tense with a verb of action, indefinite determiner phrases like “a dolphin” are usually read as universal (a kind of generic reading – “a dolphin whistles [will whistle]”). When used with a verb of state, they are usually read as existential (“a dolphin is [will be] in my car”). With the verb “have”, either reading seems acceptable (“a dolphin has dorsal fins”, “a dolphin has a broken tooth”). In the past tense, they may seem unambiguously existential (“a dolphin whistled”), but there are exceptions (“A telephone call cost only a dime in 1950”). In the present case, the most natural reading is as a restricted universal generalization,

$$\forall x(Kx \rightarrow Fx)$$

An interesting exception to this reading occurs in the narrative present -- the style of play-by-play sports reporting, “A halfback rolls out to the left. He feints to the right. ...” So a play by play reporter at Sea World might report, “A dolphin jumps. It touches the pole. ...”, yielding,

$$\exists x(Kx \wedge Fx)$$

Are there other uses that would yield the existential reading for this sentence?

3.215 Only a dog is long-haired.

Using the suggestion of 2, we can transform 15 into the stylistic variant,

“nothing distinct from a dog is long-haired”,
and then to,

$\sim \exists x(x \text{ is not a dog} \wedge x \text{ is long-haired}),$

which then becomes

$\sim \exists x(\sim Ix \wedge Mx)$

3.216 Unless Spot is a canary, he neither sings nor is yellow.

Here the pronoun can be treated as a mere shorthand. Thus, “unless” can be regarded as the main connective, as in 10. If we regard the unless clause as subordinate, which is strictly speaking correct, we might reverse the order of the disjuncts (though either order is acceptable) yielding,

“Spot neither sings nor is yellow unless Spot is a canary”,
which would be symbolized, using our convention for “neither/nor” (see 5), as

$\sim(Ga \vee La) \vee Ja$

3.217 Both Spot and Kiwi jump, but only if they are pets are they petted.

The interest in this problem stems from the two occurrences of the pronoun “they” in the second conjunct,

“only if they are pets are they petted”.

If we treat both as shorthands, we would replace each of them with its antecedent to form,

“only if both Spot and Kiwi are pets are both Spot and Kiwi petted”.

This is a familiar form, so we can begin the symbolization as,

(both Spot and Kiwi petted $\rightarrow$ both Spot and Kiwi are pets).

Here the antecedent and consequent are obviously conjunctions, yielding,

(Spot is petted and Kiwi is petted $\rightarrow$ Spot is a pet and Kiwi is a pet).

The whole of 17 is thus symbolized as,

$[(Fa \wedge Fb) \wedge ((Ha \wedge Hb) \rightarrow (Na \wedge Nb))]$

On the other hand, if we could treat the second as a shorthand, but the first differently, we could get,

“only if they are pets are Spot and Kiwi (each) petted”,

which becomes,

“Spot and Kiwi are (each) petted only if they are pets”,

which could be parsed as,

“[Spot and Kiwi] [are (each) petted only if they are pets]”,

which becomes the conjunction,

(Spot is petted only if he is a pet $\wedge$ Kiwi is petted only if he is a pet).

This yields, for the original sentence,

$[(Fa \wedge Fb) \wedge ((Ha \rightarrow Na) \wedge (Hb \rightarrow Nb))]$

Is this last a plausible reading of 17?

ADDED 218 Neither dolphins nor canaries are long-haired.

This is again, as in 201, a sentence with a ‘sententially’ compound subject. As in 201, it is treated as a sentential compound. Following our policy of treating “neither/nor” as “it is not the case that (either/or)” leads us to,

“It is not the case that (either dolphins or canaries are long-haired)”,
which quickly yields,

“It is not the case that (dolphins are long-haired or canaries are long-haired)”

The bare plurals in “dolphins are long-haired” and “canaries are long-haired” are treated as universals, although this is probably not quite correct,

$$\sim(\forall x(Kx \rightarrow Mx) \vee \forall x(Jx \rightarrow Mx))$$

which is equivalent to

$$\sim \forall x(Kx \rightarrow Mx) \wedge \sim \forall x(Jx \rightarrow Mx).$$

Note that

$$\sim \exists x(Kx \wedge Mx) \wedge \sim \exists x(Jx \wedge Mx)$$

says something quite different.

ADDED 219 Only yellow canaries and long-haired dogs can fly.

Note that this sentence is ambiguous as between, “*Among canaries* only the yellow ones can fly, and *among dogs*, only the long-haired can fly” and “*Whatever can fly* is either a yellow canary or a long-haired dog.”

ADDED 220 Only if a dolphin is petted does it jump.

This sentence is ambiguous between an existential (try making it “a certain dolphin”) and a universal reading of “a dolphin”. So we have both:

Dolphins jump only if petted

$$\forall x(Kx \rightarrow (Fx \rightarrow Hx)),$$

and

Some dolphin jumps only if petted

$$\exists x(Kx \wedge (Fx \rightarrow Hx))$$

ADDED 221a-g The following are equivalent, and perhaps stylistic variants of one another.

Any dog that is long-haired can fly.

A dog that is long-haired can fly.

If a dog is long-haired, it can fly.

A dog, if it is long-haired, can fly.

A dog, if long-haired, can fly.

Dogs, if they are long-haired, can fly.

Dogs, if long-haired, can fly.

ADDED 222 Dogs can fly, but only if they are long-haired.

This sentence is puzzling. It seems natural and idiomatic. But from a logical point of view it seems almost contradictory. Perhaps the second clause *is* a hedge, taking back some of what was asserted in the first clause. Or perhaps the first clause should be read as an existential rather than a universal. All in all, I am inclined to read the “but only if” construction (as opposed to plain “only if”) as “if and only if”. Thus,

$$\forall x(Ix \rightarrow (Ox \leftrightarrow Mx))$$

ADDED 223 Dolphins jump if petted, but not if not petted.

$$\forall x(Kx \rightarrow (Hx \rightarrow Fx)) \wedge \forall x(Kx \rightarrow (\sim Hx \rightarrow \sim Fx))$$

which is equivalent to

$$\forall x((Kx \rightarrow (Hx \rightarrow Fx)) \wedge (Kx \rightarrow (\sim Hx \rightarrow \sim Fx)))$$

which is equivalent to

$$\forall x(Kx \rightarrow (Hx \rightarrow Fx) \wedge (\sim Hx \rightarrow \sim Fx))$$

which is equivalent to

$$\forall x(Kx \rightarrow (Hx \leftrightarrow Fx))$$

ADDED 224 If Feathers can fly any canary can fly, and if any canary can fly, Feathers can.

The point of this problem is to illustrate the shiftiness of “any”. (See further discussion of “any” and “a” below.) The two conjuncts differ only in that the second is the exact converse of the first, antecedent and consequent are reversed. Yet the very same sentence, “any canary can fly”, receives a different reading of “any” when it appears in the antecedent from that which it receives when it appears in the consequent.

$$(Oc \rightarrow \forall x(Jx \rightarrow Ox)) \wedge (\exists x(Jx \wedge Ox) \rightarrow Oc)$$

[It may be that there is a secondary reading of the second conjunct according to which that “any” is also read as universal, but even so, the ambiguity appears *only* in the second conjunct, again illustrating the extreme context sensitivity of “any”.]

ADDED 225 Spot and Kiwi each jump if they are petted.

This sentence is ambiguous between, “Each jumps if *it* is petted” and “Each jumps if *both* are petted.”

Chapter V Symbolizations – Examples and Explanations

Statements Of Number

The most fundamental forms that are symbolized in the language of Chapter V are *statements of number*. All have many equivalent forms, but we will fix on certain standard forms.

There is at least one thing.

$\exists x x=x$ (This is a theorem.)

There is at most one thing.

$\forall x \forall y (x=y)$

There is exactly one thing.

$\exists x \forall y y=x$

There is at least one F.

$\exists x Fx$

There is at most one F.

$\forall x \forall y (Fx \wedge Fy \rightarrow x=y)$

There is exactly one F.

$\exists x \forall y (Fy \leftrightarrow y=x)$

At least one F is G.

$\exists x (Fx \wedge Gx)$

At most one F is G.

$\forall x \forall y (Fx \wedge Gx \wedge Fy \wedge Gy \rightarrow x=y)$

Exactly one F is G

$\exists x \forall y (Fy \wedge Gy \leftrightarrow y=x)$

There are at least two things.

$\exists x \exists y x \neq y$

There are at most two things.

$\forall x \forall y \forall z (x=y \vee x=z \vee y=z)$

There are exactly two things.

$\exists x \exists y (x \neq y \wedge \forall z (z=x \vee z=y))$

There are at least two Fs.

$\exists x \exists y (Fx \wedge Fy \wedge x \neq y)$

There are at most two Fs.

$\forall x \forall y \forall z (Fx \wedge Fy \wedge Fz \rightarrow x=y \vee x=z \vee y=z)$

There are exactly two Fs.

$\exists x \exists y (x \neq y \wedge \forall z (Fz \leftrightarrow z=x \vee z=y))$

At least two Fs are G.

$\exists x \exists y (Fx \wedge Gx \wedge Fy \wedge Gy \wedge x \neq y)$

At most two Fs are G.

$\forall x \forall y \forall z (Fx \wedge Gx \wedge Fy \wedge Gy \wedge Fz \wedge Gz \rightarrow x=y \vee x=z \vee y=z)$

Exactly two Fs are G.

$$\exists x \exists y (x \neq y \wedge \forall z (Fz \wedge Gz \leftrightarrow z = x \vee z = y))$$

There are at least three things.

$$\exists x \exists y \exists z (x \neq y \wedge x \neq z \wedge y \neq z)$$

There are at most three things.

$$\forall x \forall y \forall z \forall w (x=y \vee x=z \vee x=w \vee y=z \vee y=w \vee z=w)$$

There are exactly three things.

$$\exists x \exists y \exists z (x \neq y \wedge x \neq z \wedge y \neq z \wedge \forall w (w=x \vee w=y \vee w=z))$$

There are at least three Fs.

$$\exists x \exists y \exists z (Fx \wedge Fy \wedge Fz \wedge x \neq y \wedge x \neq z \wedge y \neq z)$$

There are at most three Fs.

$$\forall x \forall y \forall z \forall w (Fx \wedge Fy \wedge Fz \wedge Fw \rightarrow x=y \vee x=z \vee x=w \vee y=z \vee y=w \vee z=w)$$

There are exactly three Fs.

$$\exists x \exists y \exists z (x \neq y \wedge x \neq z \wedge y \neq z \wedge \forall w (Fw \leftrightarrow w=x \vee w=y \vee w=z))$$

Note the following equivalences:

There are at least $n+1$ Fs.

At least $n+1$ Fs are G

There are at most n Fs.

At most n Fs are G

There are exactly n Fs.

Exactly n Fs are G.

There are less than $n+1$ Fs.

Less [fewer] than $n+1$ Fs are G

No less than n Fs are G

There are more than n Fs.

More than n Fs are G

No more than n Fs are G

It is not the case that there are at most n Fs.

It is not the case that at most n Fs are G.

It is not the case that there are at least $n+1$ Fs.

It is not the case that at least $n+1$ Fs are G.

There are at least n Fs, and there are at most n Fs.

At least n Fs are G, and at most n Fs are G.

There are at most n Fs.

At most n Fs are G

At least n Fs are G.

There are at least $n+1$ Fs.

At least $n+1$ Fs are G.

At most n Fs are G.

These equivalences provide for multiple symbolizations of numerical determiner phrases. But note that there are standard ways for symbolizing “at most n ”, “at least n ”, and “exactly n ”. On exams, we accept any answer equivalent to a correct answer, but the “correct” answers entered into the program tend to follow the standards.

A typical, complex, symbolization problem will involve several numerical determiner phrases. For example, At most two students consulted at least two instructors for help with exactly one problem. A significant difficulty in doing such exercises is determining the relative scopes of the three determiner phrases. For purposes of our class, we will add an explicit version of Stipulation 3:

Stipulation 5: Unless it is otherwise signified in the problem, we will assume that the left to right order of the determiner phrases reflects the scopes of the underlying logical form.

Thus, in the example above, “At most two students” would have primary scope, “at least two instructors” secondary scope, and “exactly one problem” would have tertiary scope.

We know from the example, “A fool is born every minute”, that, in fact, the left to right order does not invariably reflect the intended interpretation of a sentence containing more than one determiner phrase. That is why we need to make a stipulation!

Here are some worked out examples:

At most one Junior derided at least two Freshman.

$$\forall x \forall y ((Jx \wedge (x \text{ derided at least two freshman})) \wedge (Jy \wedge (y \text{ derided at least two freshman})) \rightarrow x=y)$$

$$\forall x \forall y ((Jx \wedge \exists z \exists w (Fz \wedge Fw \wedge z \neq w \wedge R(xz) \wedge R(xw))) \wedge (Jy \wedge \exists z \exists w (Fz \wedge Fw \wedge z \neq w \wedge R(yz) \wedge R(yw)))) \rightarrow x=y)$$

Exactly one Junior derided exactly one Freshman.

$$\exists x \forall y (Jy \wedge y \text{ derided exactly one Freshman} \leftrightarrow y=x)$$

$$\exists x \forall y (Jy \wedge \exists z \forall w (Fw \wedge R(yw) \leftrightarrow w=z) \leftrightarrow y=x)$$

Note that the above says something quite different from what it would say if we gave “exactly one Freshman” primary scope. It would then say: Exactly one Freshman was derided by exactly one Junior.

$$\exists z \forall w (Fw \wedge w \text{ was derided by exactly one Junior} \leftrightarrow w=z)$$

$$\exists z \forall w (Fw \wedge \exists x \forall y (Jy \wedge R(yw) \leftrightarrow y=x) \leftrightarrow w=z)$$

Can you see intuitively that “Exactly one Freshman was derided by exactly one Junior” is not implied by “Exactly one Junior derided exactly one Freshman”? [Hint: Perhaps the other Juniors derided many Freshman.] You may wish to try to construct a model in which the latter (the original sentence) is true but the former is false.

No Junior derided less than two Freshman. [Recall: “less than two” = “at most one”, when dealing with wholes.]

$$\sim \exists x (Jx \wedge x \text{ derided less than two Freshman})$$

$$\sim \exists x (Jx \wedge \forall y \forall z (Fy \wedge R(xy) \wedge Fz \wedge R(xz) \rightarrow y=z))$$

Exactly one Junior derided no Freshmen, and that Junior is to be commended.

$$\exists x (\forall y (Jy \wedge \sim \exists z (Fz \wedge R(yz)) \leftrightarrow y=x) \wedge Nx)$$

Note that

$$(\forall y (Jy \wedge \sim \exists z (Fz \wedge R(yz)) \leftrightarrow y=x)$$

says that x, and only x, is a Junior who derided no Freshman.

There is exactly one middle-aged Freshman.

$$\exists x \forall y (Fy \wedge Mx \leftrightarrow y=x)$$

There is exactly one middle-aged Freshman who is gray-haired.

[There is exactly one middle-aged, gray-haired Freshman.]

$$\exists x \forall y (Fy \wedge My \wedge Gy \leftrightarrow y=x)$$

There is exactly one middle-aged Freshman, and he or she is gray-haired.

$$\exists x (\forall y (Fy \leftrightarrow y=x) \wedge Gx)$$

[This is Russell’s symbolization of “The middle-aged Freshman is gray-haired.”]

Comparatives and Superlatives

Although there is no obvious way to express the comparative “is faster than” in term of the monadic predicate “is fast”, there *is* a way to express the superlative “is fastest” in terms of the comparative.

$S\{1\}\{2\}$: $\{1\}$ is faster than $\{2\}$

$F\{1\}$: $\{1\}$ is a Freshman

a: Alonzo

Alonzo is fastest.

$\forall x(x \neq a \rightarrow a \text{ is faster than } x)$.

Don’t forget the $x \neq a$. “ $\forall x(a \text{ is faster than } x)$ ” implies “a is faster than a”, which is absurd.

The absurdity should be reflected in logic, but it isn’t. The reason is that we have no way of indicating that a given two-place predicate is a comparative. We would need a new logical sign to do so. So instead, if we wish to capture the intuitive logical truths involving comparatives, we must add special premises to each argument involving comparatives. These premises would express the *transitivity* and the *antisymmetry* of the comparative:

$\forall x \forall y \forall z (S(xy) \wedge S(yz) \rightarrow S(xz))$

$\forall x \forall y (S(xy) \rightarrow \sim S(yx))$

Together, these two imply the *antireflexivity* of the comparative:

$\forall x \sim S(xx)$

Usually, superlatives are relativized.

Alonzo is a fastest Freshman.

$Fa \wedge \forall x (Fx \wedge x \neq a \rightarrow S(ax))$.

Of course there can be only one fastest Freshman. So *any* fastest Freshman is *the* fastest Freshman. However, to *prove* that there is at most one fastest Freshman requires antisymmetry. Try it!

Issues involving “only”.

A number of interesting issues arise in the symbolization of phrases involving “only”

“Only” can appear both with a general term, “only dogs”, and with a singular term. “only Emily”. It can also appear with compound singular terms, “only Emily and Denise”.

To begin with the general term:

Only dogs can fly.

$\forall x (Fx \rightarrow Gx)$

which might be paraphrased

Nothing other than a dog can fly

$\sim \exists x (\sim Gx \wedge Fx)$

Only long-haired-dogs can fly.

One expects

$$\forall x(Fx \rightarrow Lx \wedge Gx)$$

or, equivalently,

$$\sim \exists x(\sim(Lx \wedge Gx) \wedge Fx)$$

But this sentence is actually ambiguous as between

Nothing other than a long-haired dog can fly

$$\sim \exists x(\sim(Lx \wedge Gx) \wedge Fx)$$

and

No dog that is not long-haired dog can fly

$$\sim \exists x(Gx \wedge \sim Lx \wedge Fx)$$

which is equivalent to

$$\forall x(Gx \rightarrow (Fx \rightarrow Lx))$$

and which might be paraphrased,

Among dogs, only the long-haired can fly.

The “only” plus general term construction can occur as subject, object, or indirect object.

Only Freshmen threw Emily to Denise.

$$\forall x(T(xed) \rightarrow Fx)$$

Emily threw only Freshmen to Denise.

$$\forall x(T(xed) \rightarrow Fx)$$

Emily only threw Denise to Freshmen.

Emily threw Denise only to Freshmen.

$$\forall x(T(edx) \rightarrow Fx)$$

Only Freshmen threw Emily only to Freshmen.

$$\forall x(x \text{ threw Emily only to Freshmen} \rightarrow x \text{ is a Freshman})$$

$$\forall x(\forall y(x \text{ threw Emily to } y \rightarrow y \text{ is a Freshman}) \rightarrow x \text{ is a Freshman})$$

$$\forall x(\forall y(T(xey) \rightarrow Fy) \rightarrow Fx)$$

When numerical determiners are used with “only” a surprising change takes place. The “only” seems to take on the meaning of “exactly”. “Only one dog can fly” seems to imply not only that *no more* than one dog can fly, but that one dog *can* fly. This is equivalent to saying that exactly one dog can fly.

Only one dog can fly.

Paraphrased as: No more than one dog can fly, but one can.

$$\sim \exists x \exists y(Gx \wedge Gy \wedge x \neq y \wedge (Fx \wedge Fy)) \wedge \exists u(Gu \wedge Fu)$$

which is equivalent to our standard formulation

$$\exists x \forall y(Gy \wedge Fy \leftrightarrow x=y)$$

Only two dogs can fly.

Paraphrased as: No more than two dogs can fly, and at least two dogs can fly.

$$\sim \exists x \exists y \exists z((Gx \wedge Gy \wedge Gz) \wedge x \neq y \wedge x \neq z \wedge y \neq z \wedge (Fx \wedge Fy \wedge Fz)) \wedge \exists u \exists v((Gu \wedge Fu) \wedge (Gv \wedge Fv) \wedge u \neq v)$$

which, again, is equivalent to our standard formulation

$$\exists x \exists y(x \neq y \wedge \forall z(Gz \wedge Fz \leftrightarrow z=x \vee z=y))$$

When combined with singular terms, “only” seems to behave in the way it does when combined with numerical determiners. “Only Emily enrolled” certainly implies that no one other than Emily enrolled, but it also seems to imply that Emily *did* enroll.

Only Emily enrolled.

Paraphrased as: Nothing other than Emily enrolled, but Emily *did* enroll.

$\sim \exists x(x \neq e \wedge Nx) \wedge Ne$

which is equivalent to our standard formulation of
Emily and only Emily enrolled

$\forall x(Nx \leftrightarrow x=e)$

Only Emily and Denise enrolled.

$\forall x(Nx \leftrightarrow x=e \vee x=d)$

Emily envies only Denise

$\forall x(V(ex) \leftrightarrow x=d)$

Only Emily envies Denise

$\forall x(V(xd) \leftrightarrow x=e)$

Only Emily envies only Denise

$\forall x(x \text{ envies only Denise} \leftrightarrow x=e)$

$\forall x(\forall y(V(xy) \leftrightarrow y=d) \leftrightarrow x=e)$

Only Denise is envied by only Emily

$\forall y(y \text{ is envied by only Emily} \leftrightarrow y=d)$

$\forall y(\text{only Emily envies } y \leftrightarrow y=d)$

$\forall y(\forall x(V(xy) \leftrightarrow x=e) \leftrightarrow y=d)$

Are the last two equivalent?

Sentences may also mix the use of “only” with general terms and with names or numerical terms.

Starting with the logical truth:

Emily envies those that Emily envies.

$\forall x(\text{Emily envies } x \rightarrow \text{Emily envies } x)$

we can form eight variants by sprinkling “only”s in three places:

Emily envies those that Emily envies.

[Only?] Emily envies [only?] those that [only?] Emily envies.

Our treatment of “only Emily” makes it a logical truth of:

Emily envies all those that only Emily envies.

$\forall x(\text{only Emily envies } x \rightarrow \text{Emily envies } x)$

$\forall x(\forall y(V(yx) \leftrightarrow y=e) \rightarrow V(ex))$

Can you prove this?

Does our treatment also make a logical truth of:

Only Emily envies all those that only Emily envies.

$\forall z(\forall x(\text{only Emily envies } x \rightarrow z \text{ envies } x) \leftrightarrow z = \text{Emily})$

$\forall z(\forall x(\forall y(V(yx) \leftrightarrow y = e) \rightarrow V(zx)) \leftrightarrow z = e)$

Can you prove this?

Can you show it to be invalid?

Finally, we have a case in which “only” appears in all three of its possible contexts: with a numerical determiner, with a name, and with a general term (here, for stylistic reasons, we write “only to Juniors” rather than “to only Juniors”).

Only two Freshmen threw only Emily only to Juniors.

Following our conventions regarding “only” with names and numerical determiners, this becomes:

Exactly two Freshmen threw (Emily and only Emily) only to Juniors.

Following our convention to take the scopes in left to right order, this becomes:

$\exists x \exists y(x \neq y \wedge Fx \wedge Fy \wedge \forall z(z \text{ is a Freshmen who threw [Emily and only Emily] only to Juniors} \leftrightarrow z = x \vee z = y))$

$\exists x \exists y(x \neq y \wedge Fx \wedge Fy \wedge \forall z(Fz \wedge z \text{ threw [Emily and only Emily] only to Juniors} \leftrightarrow z = x \vee z = y))$

$\exists x \exists y(x \neq y \wedge Fx \wedge Fy \wedge \forall z(Fz \wedge \forall w(z \text{ threw } w \text{ only to Juniors} \leftrightarrow w = \text{Emily}) \leftrightarrow z = x \vee z = y))$

$\exists x \exists y(x \neq y \wedge Fx \wedge Fy \wedge \forall z(Fz \wedge \forall w(\forall v(z \text{ threw } w \text{ to } v \rightarrow v \text{ is a Junior}) \leftrightarrow w = \text{Emily}) \leftrightarrow z = x \vee z = y))$

$\exists x \exists y(x \neq y \wedge Fx \wedge Fy \wedge \forall z(Fz \wedge \forall w(\forall v(T(zwv) \rightarrow Jv) \leftrightarrow w = e) \leftrightarrow z = x \vee z = y))$

Because there may be individual differences in how phrases involving “only” are read, let us make stipulations to determine what will count as correct in our class.

Stipulation 6: For purposes of our class, we will regard “only” with a numerical determiner as having the sense of “exactly”, and we will regard phrases like “only Emily” as synonymous with “Emily and nothing other than Emily”. [Note that these stipulations do not affect the use of “only” with a general term like “dogs” or “those who admire dogs”] [Also note that we have carefully avoided saying the personal pronoun in “Emily and no *one* other than Emily” so as to avoid having to comply with Stipulation 4.]

DK